

Exploratory Analysis, KPI Development and Business Insights

1. INTRODUCTION

HealthConnect Clinic is seeking to better understand patient appointment attendance patterns and identify factors associated with missed appointments. Week 5 focused on exploratory analysis, KPI development and the identification of business insights that could support improved appointment attendance and operational planning.

The analysis was conducted using the HealthConnect appointment dataset containing 5,000 appointment records across 18 variables. Power Query was used for data preparation and Power BI was used for exploratory analysis, KPI development and data visualisation.

2. DATASET OVERVIEW

The HealthConnect dataset contains 5,000 appointment records representing 1,696 unique patients across 18 variables.

The dataset includes information relating to:

- Patient characteristics
- Appointment characteristics
- Appointment outcomes
- Booking lead time
- Previous appointment history
- Previous no-show behaviour
- Reminder status and reminder channel
- Distance to the clinic
- Waiting time

The recorded appointment outcomes were:

Outcome	Count	Percentage
No-Show	2,423	48.5% of all records
Attended	2,314	46.3% of all records
Cancelled	263	5.3% of all records
Total	5,000	100%

For the no-show and attendance KPIs, cancelled appointments were excluded from the denominator because a cancellation is different from a patient failing to attend an appointment.

3. KPI DEVELOPMENT

Five KPIs were developed to measure appointment performance and support the business questions defined during the earlier analysis stage.

KPI	Result	Business Relevance
Total Appointments	5,000	Measures overall appointment volume.
No-Show Rate	51.2%	Measures the proportion of non-cancelled appointments that were missed.
Attendance Rate	48.8%	Measures the proportion of non-cancelled appointments attended.

Cancellation Rate	5.3%	Measures the proportion of all appointments that were cancelled.
Average Booking Lead Time	29.6 days	Shows the average time between appointment booking and the scheduled appointment.

The KPI results show that no-shows represent a major operational challenge for HealthConnect. More than half of the non-cancelled appointments in the dataset resulted in a no-show.

4. Exploratory Analysis

The analysis examined appointment outcomes across patient characteristics, appointment characteristics, previous appointment behaviour, reminder activity, booking lead time, distance and waiting time.

The Power BI dashboard consists of two analytical pages: an Executive Overview and a No-Show Drivers page.

5. KEY BUSINESS INSIGHTS

Insight 1 - No-shows represent a significant operational challenge

The overall no-show rate was 51.2%, compared with an attendance rate of 48.8% among non-cancelled appointments.

This indicates that missed appointments are not a minor issue within the dataset. More than half of appointments that were either attended or missed resulted in patients not attending.

Business implication

A high no-show rate may contribute to unused appointment capacity and reduce the efficiency of clinic scheduling. Reducing missed appointments could therefore improve the utilisation of available appointment slots.

Insight 2 - Previous no-show behaviour is strongly associated with current no-shows

The analysis showed a clear increase in no-show rates as previous no-show history increased:

Previous No-Show Group	No-Show Rate
No Previous No-Shows	46.3%
1 Previous No-Show	55.9%
2+ Previous No-Shows	63.5%

Patients with two or more previous no-shows had a current no-show rate of approximately 63.5%, compared with 46.3% among patients with no previous no-shows.

Business implication

Previous attendance behaviour may provide a useful indicator for identifying appointments that require additional confirmation or engagement.

Insight 3 - Longer booking lead times are associated with higher no-show rates

A strong relationship was observed between booking lead time and no-show behaviour.

Booking Lead Time	No-Show Rate
0–3 Days	26.6%
4–7 Days	32.3%
8–14 Days	35.2%
15–30 Days	45.5%
31+ Days	63.9%

Appointments booked 31 or more days in advance had a no-show rate of approximately 63.9%, compared with 26.6% for appointments booked within 0–3 days.

Business implication

Longer booking periods may provide more opportunity for plans to change or appointments to be forgotten. HealthConnect could therefore consider additional confirmation strategies for appointments booked far in advance.

Insight 4 - Reminders are associated with higher attendance

Attendance rates differed according to reminder status.

Reminder Status	Attendance Rate
Reminder Sent	50.1%
No Reminder	45.4%

Appointments associated with a reminder had an attendance rate of approximately 50.1%, compared with 45.4% for appointments without a reminder.

This represents a difference of approximately 4.8 percentage points.

Business implication

The result suggests that reminder coverage may be a useful area for operational improvement. However, the analysis shows an association rather than proving that reminders directly caused the difference in attendance.

Insight 5 - Greater distance from the clinic is associated with higher no-show rates

No-show rates varied across distance groups.

Distance Group	No-Show Rate
0–5 km	48.7%
5–10 km	49.3%
11–20 km	52.3%
21+ km	60.5%

Patients in the 21+ km group had the highest observed no-show rate at approximately 60.5%, compared with 48.7% among patients within 0–5 km.

Business implication

Distance may represent an accessibility-related factor associated with missed appointments. Further investigation could help determine whether patients travelling longer distances require different scheduling or support options.

6. SUPPORTING FINDINGS

Additional differences were observed across appointment characteristics.

Follow-up appointments had the highest no-show rate among the appointment types at approximately 54.2%, followed by Diagnostic Tests at approximately 52.0%.

No-show rates also varied by appointment time, with evening appointments showing approximately 52.7%, compared with 51.2% for afternoon appointments and 50.7% for morning appointments.

Differences across age groups and appointment days were relatively smaller than the patterns observed for previous no-show history and booking lead time.

These variables should therefore continue to be monitored, but they were not considered as strong headline drivers as the patterns identified above.

7. BUSINESS RECOMMENDATIONS

Recommendation 1 - Introduce targeted follow-up for patients with previous no-shows

HealthConnect should consider identifying patients with previous missed appointments and applying additional confirmation or engagement measures before future appointments.

Patients with repeated previous no-shows recorded a substantially higher current no-show rate, making this group particularly relevant for targeted intervention.

Recommendation 2 - Strengthen confirmation for appointments booked far in advance

Appointments booked more than 31 days ahead recorded a no-show rate of approximately 63.9%.

HealthConnect could consider additional reminder or confirmation points for appointments scheduled several weeks in advance rather than relying only on a single reminder close to the appointment date.

Recommendation 3 - Review reminder coverage

Appointments associated with reminders had a higher attendance rate than appointments without reminders.

HealthConnect should review appointments that currently do not receive reminders and assess whether broader reminder coverage could improve attendance.

Future analysis should evaluate reminder timing and patient characteristics before implementing changes at scale.

Recommendation 4 - Consider accessibility for patients travelling longer distances

Patients travelling 21+ km had the highest observed no-show rate.

HealthConnect could investigate whether transportation difficulties, travel time or other accessibility factors contribute to missed appointments. Where clinically appropriate, options such as flexible scheduling or remote consultation could be considered.

Recommendation 5 - Use appointment history as an operational risk indicator

Previous no-show behaviour could be incorporated into an appointment monitoring process.

Rather than treating all appointments identically, HealthConnect could prioritise additional confirmation for patients with repeated previous no-shows.

8. ASSUMPTIONS

The following assumptions were applied during the analysis:

- Each appointment ID represents one appointment record.
- Multiple records associated with the same patient ID represent legitimate repeat appointments rather than duplicate records.
- Cancelled appointments were treated separately from attended and no-show appointments.
- No-show and attendance rates were calculated using attended and no-show appointments as the denominator.
- Missing reminder-channel values were interpreted as representing appointments for which no reminder channel was recorded.
- Missing distance and waiting-time values were retained as missing rather than artificially imputed.
- Observed relationships were interpreted as associations and not as proof of causation.

9. LIMITATIONS

The analysis has several limitations:

- The dataset does not provide a direct explanation for why patients miss appointments.
- Relationships identified in the analysis do not establish causation.
- Some variables contain missing values.
- Certain groups contain different numbers of observations, which may affect the stability of their rates.
- The dataset represents a specific set of HealthConnect appointment records and may not fully represent future patient behaviour.
- Booking-date consistency requires further validation before relying heavily on date-based time-series analysis.

10. RISKS AND DEPENDENCIES

Key risks and dependencies identified during the analysis include:

- Incorrect interpretation of appointment and booking dates could affect time-based analysis.
- Missing values may influence comparisons involving distance, waiting time and reminder information.
- Small subgroups may produce less stable rates.
- Operational recommendations should be validated with HealthConnect stakeholders before implementation.
- Additional analysis may be required to determine whether observed associations remain consistent across different patient groups.

11. CROSS-TRACK COLLABORATION

Cross-Track Collaboration - Product/Business Analysis

A simulated cross-track discussion was considered with the Product/Business Analysis track to review the operational implications of the appointment attendance findings. The discussion focused on how factors such as previous no-show history, booking lead time, reminder status, and distance from the clinic could influence appointment management and patient engagement.

The collaboration helped connect the analytical findings to potential operational actions, particularly the need for targeted follow-ups for patients with previous no-shows, improved reminder coverage, and additional attention to appointments booked far in advance or for patients travelling longer distances.

This perspective strengthened the analysis by moving beyond identifying patterns in the data and considering how the findings could support practical improvements in appointment management.

12. CONCLUSION

The Week 5 analysis identified several important patterns in HealthConnect's appointment data. The overall no-show rate of 51.2% indicates a significant attendance challenge.

The strongest observed patterns were associated with previous no-show history, booking lead time, reminder status and distance to the clinic. In particular, patients with two or more previous no-shows recorded a no-show rate of 63.5%, while appointments booked 31 or more days in advance recorded a no-show rate of 63.9%.

These findings provide a foundation for targeted appointment confirmation, improved reminder strategies and further investigation of accessibility-related barriers.

The results should be treated as exploratory associations rather than causal relationships. Further validation and deeper analysis will be required before operational changes are implemented.